# Choosing The BEST SMARTPHONE Under ₹ 10,000

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If you want a smartphone with the best of features but not costing more than ₹ 10,000, this guide can help you get the best value for your money. It explains what the different smartphone specifications represent and what's available from various brands.

## Processor

Processor makes your smartphone run the functions you want. It consists of multiple units called 'cores.' The more the number of cores, the better the work division the processor can carry out. As a result, it becomes easier for the phone to run multiple applications at a time without any hiccups.

The smartphone's performance also depends on the processor's clock speed, which is measured in gigahertz (GHz). This, in turn, utilises the number of bits labelled on the processor to perform a function in a given time. Processors are usually 32-bit or 64-bit—the higher the bit count, the more the data it can handle at a clock beat. Again, the higher the clock speed, the more the actions utilising the bits. So, for example, a 2.4GHz octa-core processor works better than a 1.2GHz octa-core processor, and a 2.4GHz 64-bit chip is better than a 2.4GHz 32-bit chip.

In the market, you will find smartphones with different processor options, the majority of which use Snapdragon and Mediatek. Mediatek is an adequate processor choice for low-budget phones, given its low power consumption and high load capacity at reasonable price. Snapdragon chipsets are usually used in comparatively costlier phones. Improved performance and efficiency make Snapdragon a good catch. Quite a few low-budget phones incorporate Spreadtrum processors. System-on-chip technology brings the whole system—including the graphic processing unit (GPU)—to the processor. Snapdragon processors come with Adreno GPU, while Mediatek or Exynos processors carry a Mali unit. GPU performance can be judged from its version and benchmark tests.


A Mediatek MT6589 quadcore processor for base-level smartphones (Image courtesy: flickr)

## RAM

Random access memory (RAM) is responsible for the ease of information retrieval—eliminating for the system the need to access the storage or SD card to run an application. RAM holds the running as well as background apps for quick access when the app is run the next time. Consequently, it greatly eases the load on the system, allowing the phone to run smoother.

Operating system (OS) and user interface (UI) occupy a big chunk of the RAM. The remaining space is used by applications to work. So, the bigger the RAM, the greater will be the capacity to perform applications. Look out for a phone with at least 2GB RAM.

## Display

Display brightness is important for a vibrant and realistic viewing experience as well as comfortable viewing outdoors. LCD screens have been a long-time reliable companion for smartphones. The new AMOLED displays offer better colour contrasts. With recent upgrades, these are now readable in sunlight too.

The resolution of a smartphone is determined not only by its pixel measurement but also by its screen size—combining which you can get an idea of the pixel density, that is, the number of pixels fitting per square inch of the phone screen. Higher pixel density is desirable to get a better image. Generally, low-end smartphones have either HD (1280x720-pixel) or Full HD (1920x1080-pixel) resolution. Screen size of 12.7cm (5 inches) is the best fit if you want the phone to be compact yet comfortable to use.

## Battery life

Don't settle for a phone with battery capacity